THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of the proposition “If it is Sunday, then the school is closed.” is: 1

(a) The school is closed on Sunday.
(b) If it is not Sunday, then the school is not closed.
(c) If the school is not closed, then it is not Sunday.
(d) If the school is closed, then it is Sunday.

2. Which of the following is a counterexample to the proposition “If a number is divisible by 3, then it is divisible by 9.”? 1

(a) 45 **(b)** 12 **(c)** 9 **(d)** 27

3. “If a number is divisible by both 3 and 5, then it is divisible by 15.” Its converse “If a number is divisible by 15, then it is divisible by both 3 and 5” is: 1

(a) false, since 15 is odd **(b)** true
(c) false, since 30 is divisible by 15 **(d)** neither true nor false

4. A triangle has sides 6 cm, 8 cm and 11 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) right-angled, since $a^2 + b^2 = c^2$
(b) not right-angled, since $a^2 + b^2 \neq c^2$
(c) right-angled, since the sides are in increasing order
(d) right-angled, since $a + b > c$

5. The number 48 has: 1

(a) 11 factors; it is a perfect square **(b)** 10 factors; it is not a perfect square
(c) 9 factors; it is a perfect square **(d)** 12 factors; it is not a perfect square

6. A group of 40 students has an average of 60 marks and another group of 10 students has an average of 85 marks. The average of all 50 students is: 1

(a) 64 **(b)** 66 **(c)** 67 **(d)** 65

7. The averages of 3 classes of 30 students each are 40, 52, 46. The overall average is: 1

(a) 47 **(b)** 48 **(c)** 46 **(d)** 45

8. 250 mL of a solution with 20% salt is mixed with 750 mL of a solution with 12% salt. The concentration of salt in the mixture is: 1

(a) 15% **(b)** 16% **(c)** 13% **(d)** 14%

9. 100 units of a mixture with 60% metal is mixed with a mixture of 40% metal to get 50% metal. The quantity of the second mixture needed is (same units): 1

(a) 100 **(b)** 120 **(c)** 200 **(d)** 80

10. Marks of 45, 60, 90 are combined with weights 4 : 5 : 6. The weighted average is: 1

(a) 67 **(b)** 70 **(c)** 68 **(d)** 69

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give a real-life example of a proposition that is true but whose converse is false. 2

12. Test using the converse of the Baudhāyana–Pythagoras theorem whether a triangle with sides 9 cm, 40 cm, 41 cm is right-angled. 2

13. Compute the combined percentage of profit when ₹2000 of sales give 20% profit and ₹3000 of sales give 30% profit. 2

14. Two equal glasses have 10% and 20% jaggery lemonade. Without calculation, what is the jaggery percentage after mixing? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Check whether the numbers $n = 1, 2, 3, 4, 5$ make $n^2 - n + 41$ prime. Does this prove that it is always prime? Find an n that gives a counterexample. 3

16. A drink needs 8% fruit pulp. How many litres of 20% pulp concentrate should be mixed with 6 litres of 5% pulp drink to get 8%? 3

17. Write the converse of "If $x = y$, then $a + x = a + y$". Show that both are true and say where this is used. 3

OR

If n and $n + 3$ have no common factor, then n is not a multiple of 3. Write the converse. Is the converse true? 3

18. Rashi scores 60, 70, 80 and Keerthi 70, 60, 75 in strength, flexibility and agility with weights 3 : 4 : 5. Find their final marks and say who is ahead. 3

OR

A restaurant gets food, ambience and service ratings with weights 6 : 5 : 4. If the three average ratings are 4.2, 3.4 and 2.8, find the overall rating. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain 5

(i) why a proposition and its converse can both be true, using the statement about equal angles opposite equal sides in a triangle, and

(ii) the example of both being false.

OR

Case study: The statement "If a number is divisible by 6, then it is divisible by 3" is checked on the numbers 18, 24, 30. 5

(i) Verify.

(ii) Write the converse.

(iii) Test it on 9, 15, 21, 12 and find which of these are counterexamples.

(iv) What fraction of the tested numbers is divisible by 6?

20. **Case study:** A salt lake has water of 25% salinity and a river has water of 0.5% salinity. 5

(i) Find the salinity when 1 litre of lake water is mixed with 4 litres of river water.

(ii) Can a mixture with 3% salinity be made? Find the quantity of river water per litre of lake water.

(iii) Can a mixture with 30% salinity be made? Why?

OR

Case study: A teacher gives two options for the final grade: Option 1 uses the plain average of project (80), test (60) and exam (70). Option 2 uses weights 1 : 2 : 5. 5

(i) Find both.

(ii) Which option favours a student who is strong in the exam?

(iii) Name the average used in Option 2.

END OF QUESTION PAPER — Total: 40 Marks